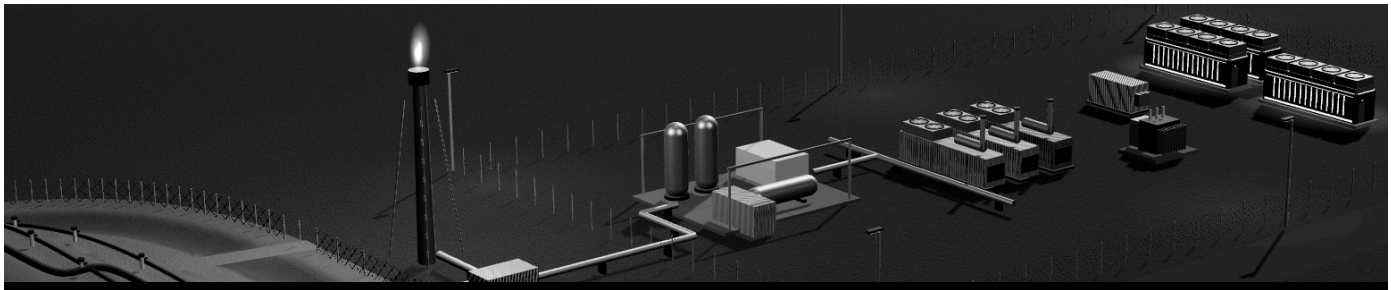
T.E. Maxson Wastewater Treatment Facility, Memphis, Tennessee: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

2.6 MW	3	8 ac	235k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $176k	$8k to $16k	$8k to $193k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=950&CH4=49

THE MATH

Lagoon biogas flared (2018 onward)	950 scfm
Methane (from about 500 Btu/cf)	49%
Energy in the gas	28.3 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 2.6 MW
Compute (power less cooling and losses, /1.2)	about 2.1 MW
Compute modules (about 1 MW each)	3
Gas engines (running plus one spare)	3 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 8 acres
Gas bought per year (95% run time)	235k MMBtu
Gas payment, up to $0.75/MMBtu	up to $176k
Land lease at $1,000 to $2,000 per acre per year	$8k to $16k

Biogas: methane at about 1,012 Btu per cubic foot. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Whether the existing biogas offtake contract is still in force
- High H2S: gas cleanup sized for it
- City data center rules (Ordinance 5986, October 2026)

Source: Shelby County Health Department, Title V renewal statement of basis (2024): about 500 MMcf a year of lagoon biogas since 2018 (about 950 scfm) at about 500 Btu/cf, all to five flares since the pipeline offtake stopped in 2017. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.